

TITLE: Post-Operative ICU Management for the Prevention of Spinal Cord Ischemia in High-Risk
Cardiovascular and Vascular Aortic Surgery Patients

PROTOCOL: Spinal Cord Ischemia Prevention

PURPOSE: Due to the complexity of aortic surgery and the vascular anatomy of the anterior spinal cord, historically, patients undergoing aortic surgery have experienced anterior spinal cord ischemia at rates greater than 10%. The University of Florida has been able to decrease rates of permanent paralysis to less than 3% by implementing a protocol designed to decrease the risk of spinal cord ischemia after aortic surgery. The continuation and updates of this protocol aim to further decrease the risk of post-operative spinal cord ischemia while also maintaining safe pre-operative practices and limiting the risks associated with the prevention therapeutics.

APPROVED:



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DEFINITIONS:

- A. MAP – mean arterial pressure
- B. SCI – spinal cord ischemia
- C. SBP – systolic blood pressure
- D. HTN – hypertension
- E. PRBC – packed red blood cells
- F. SC – subcutaneous
- G. CSF – cerebral spinal fluid
- H. APS – acute pain service

CORE PROCEDURE:

1. POST-OPERATIVE SPINAL CORD ISCHEMIA PREVENTION

- a. Hourly neurologic exam by nurse, which includes bilateral leg lift **OFF THE BED**.
 - i. Baseline neuro exam, MAP goal, hemoglobin goal, spinal drain instructions (if present) clearly communicated in sign out from surgery team.
 - ii. Surgical critical care provider will document exam on arrival from OR and then at least every shift.
- b. Permissive hypertension:
 - i. Minimum MAP per surgeon based on individual patient's risk for SCI (usual goal MAP > 90)
 - ii. Goal SBP 120-180 (MAP goal takes precedence)
 - iii. Treat HTN with IV beta blockers, clevidipine or nicardipine. NO nitroprusside, NO hydralazine, NO milrinone
 - 1. Discuss risks v benefits with surgeon prior to starting Milrinone
 - iv. Norepinephrine (or phenylephrine) infusion may be used to maintain MAP goal. If unable to reach MAP goal with norepinephrine 0.1mcg/kg/min (or

phenylephrine 1mcg/kg/min), contact surgery fellow/attending about lowering MAP goal or adding second pressor.

- c. Transfuse PRBC to maintain goal hemoglobin > 9mg/dL for first 5 days post-op.
- d. SPINAL DRAIN MANAGEMENT (directed by Surgery fellow/attending)
 - i. Zero at phlebostatic axis. Set at 10mmHg (unless otherwise specified). Confirm with OR team on patient arrival in ICU.
 - ii. Immediate postop: spinal drain OPEN at 10mmHg; goal output: maximum of 10mL CSF/hour. If 10mL output reached prior to end of 1 hour, clamp drain and reopen at start of next hour. Patient remains on bedrest.
 - iii. If no symptoms of SCI after 48 hours: CLAMP spinal drain when determined by the surgeon and continue hourly neuro exams for 24 hours. Patient may mobilize with spinal drain clamped.
 - iv. If symptoms of SCI occur: Notify Surgery fellow/attending and REOPEN spinal drain. Set at 10mmHg (or lower if necessary to alleviate symptoms).
 - v. If no symptoms of SCI after spinal drain clamped for 24 hours: spinal drain removed by Acute Pain Service
 - 1. Hold SC heparin the morning of expected date of removal.
 - vi. If CSF fluid is bloody, CLAMP drain immediately and notify Surgery fellow/attending. Discuss STAT CT head to rule out intracranial bleed.
 - vii. If patient shows no symptoms of SCI after spinal drain removal, neuro checks at night may be de-escalated to every 2 hours
 - viii. Notify surgery/fellow attending prior to initiating or increasing any anticoagulation/antiplatelet agent.
- e. Continue naloxone infusion if initiated in OR
- f. For patients with a normal neurologic exam, normothermia is the temperature goal (Bair hugger permitted). If concern for SCI, only passive warming with blankets and ambient temperature to reach 37 degrees
- g. Patients may continue stress ulcer prophylaxis while in ICU.

2. RESCUE MANEUVERS IF SCI OCCURS:

- a. Document leg lift (OFF THE BED) bilaterally.
- b. Raise MAP goal to > 100 (unless contraindicated or otherwise specified by Vascular Surgery)
- c. If NO SPINAL DRAIN in place, contact APS immediately for emergent spinal drain placement.
- d. If spinal drain already in place, discuss these options with Vascular Surgery fellow or attending:
 - i. lower drain to 5mmHg
 - ii. drain a specified volume of CSF
- e. STAT CBC to document hemoglobin. Transfuse for goal Hgb > 10mg/dL.
- f. If not already infusing, start naloxone infusion at 1-1.5 mcg/kg/h through a dedicated line.
- g. Methylprednisolone 1000mg IV over 30 min (unless steroids within 12hours)
- h. Mannitol 12.5mg IV over 15 min

ASSOCIATED POLICIES:

GL-NEU-001 - Intracranial External Ventricular Drain (EVD) & Lumbar Drain (LD) Care and Maintenance Guidelines

KEYWORDS:

Spinal Cord Ischemia; Aortic Surgery; Lumbar Drain

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